

Vortex transport and screening in a closed two-dimensional Bose field

Friction, transverse force and diffusion of quantized vortices, and the polarisability of the thermal vortex-pair gas, measured in an energy-conserving classical field with no reservoir and no fit parameter

Xavier Callens

SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

October 2026 — version 2, revised after peer review

Abstract

A quantized vortex in a thermal two-dimensional Bose gas is dragged by the normal fluid (friction α), possibly pushed sideways by it (transverse coefficient α') and kicked by it (diffusion η). Whether the transverse force exists has been disputed for thirty years; the only first-principles values of α for a bosonic classical field stop at $0.18 T_{\text{BKT}}$ and carry no error bar on α' [1]; the fluctuation–dissipation relation between η and α has never been tested in a closed system and was found violated a hundredfold in one experiment [2]. We measure the three coefficients in the projected Gross–Pitaevskii equation — microcanonical, no damping term, no noise added — by tracking imprinted vortex pairs with sub-grid resolution and reading the coefficients through estimators that are exact identities of the dissipative point-vortex model (machine-checked in Lean 4). At $T/T_{\text{BKT}} = 0.14, 0.27$ and 0.43 we find $\alpha = 0.0062(4), 0.014(2)$ and $0.02(1)$, independent of pair size and of estimator, and proportional to the normal fraction, $\alpha \simeq 0.24 \rho_n/\rho$. The transverse coefficient is zero within 1% once the kinematics of a compressible pair on a torus are accounted for; the Iordanskii value ρ_n/ρ is excluded at 18σ . The vortex diffuses at 1.4–3 times the Einstein value — not 100 times — but the fluctuations are not cleanly diffusive, so the relation is left undecided by our own rule. A single pair in a closed box stops shrinking while the phonon band absorbs the impulse it sheds, a self-generated counterflow; **pending: the $L = 96$ control decides whether the stall is the box's**. On the static side we show that the depressed stiffness seen in finite-box simulations is one unscreened box-scale pair left over from relaxation, that it decays given time that grows faster than L^2 , and that the thermal vortex-pair gas has a wavevector-independent charge structure factor equal to the Kosterlitz–Thouless polarisability computed from the measured pair sizes to 5–21% — while a mode-by-mode dielectric relation between current and vortex charge, which we had registered as a prediction, fails. Every hypothesis, control and decision rule was committed before the data it governs existed, and the failures are reported with the results (Appendix A).

1 Introduction

With the normal fluid at rest, a quantized vortex of charge q moves at

$$\mathbf{v} = (1 - \alpha') \mathbf{v}_s - \alpha q \hat{z} \times \mathbf{v}_s, \quad (1)$$

where $\mathbf{v}_s$ is the superfluid velocity at its position induced by all other vortices ($\hbar = m = 1$, circulation 2π) [3, 4]. The longitudinal coefficient α is the mutual friction; α' measures the transverse force exerted by the normal component. For a weakly dissipative vortex the two sides of a long controversy predict different α' : no transverse force from the normal fluid, $\alpha' = 0$ [5],

or the Iordanskii force, $\alpha' = \rho_n/\rho$ [3, 4]. If the field around the vortex is also its heat bath, the random kicks it receives are tied to the drag by the fluctuation–dissipation (Einstein) relation $\eta = \alpha T/(2\pi\rho)$ ($k_B = 1$) [6].

For a bosonic classical field these quantities are poorly known. Shukla, Brachet and Pandit measured α in the truncated Gross–Pitaevskii equation only below $0.18 T_{\text{BKT}}$ and could not resolve the sign of α' [1]; in three-dimensional Kelvin-wave fits α' “deviates” from expectation [7]; the Einstein relation has never been tested in a closed field, helium-film analyses assume it [8], and the one cold-atom measurement of η exceeds the prediction by two orders of magnitude, attributed to trap-wall vibrations [2]. A review is [9]. A closed, energy-conserving classical field has no walls, no reservoir and no noise put in by hand: whatever drag and diffusion a vortex experiences there come from the field’s own thermal phonons.

A second, static question concerns how the same thermal field screens vortices. The superfluid stiffness is read from the transverse current response, $\rho_n(k) = \langle |J_T(k)|^2 \rangle / (TL^2)$; in the Coulomb-gas picture bound vortex pairs reduce it through their polarisation [10, 11, 12, 13]. For a compressible Bose field no functional form at finite wavevector exists: classical-field studies extrapolate the response to $k \rightarrow 0$ by fits of convenience and have reported negative superfluid fractions attributed to noise [14, 15]. We had found earlier that such negative values are carried by one non-equilibrium vortex pair of box size, and had registered as a prediction that the transverse current is, mode by mode, the point-vortex current of the detected vortex charges. That prediction is tested here and fails; what replaces it is a direct measurement of the pair gas’s polarisability from vortex positions.

The study was run as a pre-registered programme: each hypothesis, each control with its pass threshold, and each decision rule was committed to the repository before the data it governs existed, and the controls that failed are reported alongside the results. The estimators used on the data are identities of the dissipative point-vortex model; we proved them in the Lean 4 proof assistant so that what is assumed about the model is exactly what is stated (Section 2). Section 2 describes the model and the instrument, Section 3 the transport coefficients, Section 4 the behaviour of a single pair in a closed box, Section 5 the box-scale pair, equilibration and the polarisability of the pair gas, and Section 6 what is established. The protocol, with every gate and amendment, is Appendix A; the corrected vortex imprint is Appendix B; the theorems are listed in Appendix C.

2 Model, instrument and estimators

Field. We integrate the projected Gross–Pitaevskii equation $i\partial_t\psi = \mathcal{P}[-\frac{1}{2}\nabla^2\psi + g|\psi|^2\psi]$ on a square periodic box with $\hbar = m = g = 1$, mean density $n = 1$ (healing length $\xi = 1$, sound speed $c = 1$), grid spacing $\xi/2$ and a circular momentum cutoff at half the grid wavenumber; the projector $\mathcal{P}$ keeps the dynamics inside the cutoff. Energy is conserved to 10^{-7} – 10^{-6} over every run. Thermal states are prepared by relaxing a random high-energy field at the target energy and, for the warmer states, by adding phonons to a colder state and equilibrating; their temperature comes from a two-window equipartition thermometer and their superfluid fraction from the current correlators, both validated in earlier rounds of this programme; $T_{\text{BKT}}(L=64) = 0.821$ from a heating ladder. Transport runs use $L = 64$ at $T/T_{\text{BKT}} = 0.14, 0.27, 0.43$ (no thermal vortex in the base state, or fewer than 0.4 on average) and one report-only state at 0.56 with thermal pairs; the static questions use $L = 192$ at $T/T_{\text{BKT}} \approx 0.6$ – 0.85 .

Imprinting a pair. A vortex pair is written into a thermal state by multiplying the field by a theta-function phase and an amplitude factor. Two defects of the imprint used in the earlier rounds of this programme were found by the $T = 0$ control of this study and corrected (Appendix B): the theta-function phase of a single pair is periodic only up to a constant jump $2\pi d/L$ across one boundary of the box, which we remove by a uniform phase gradient; and

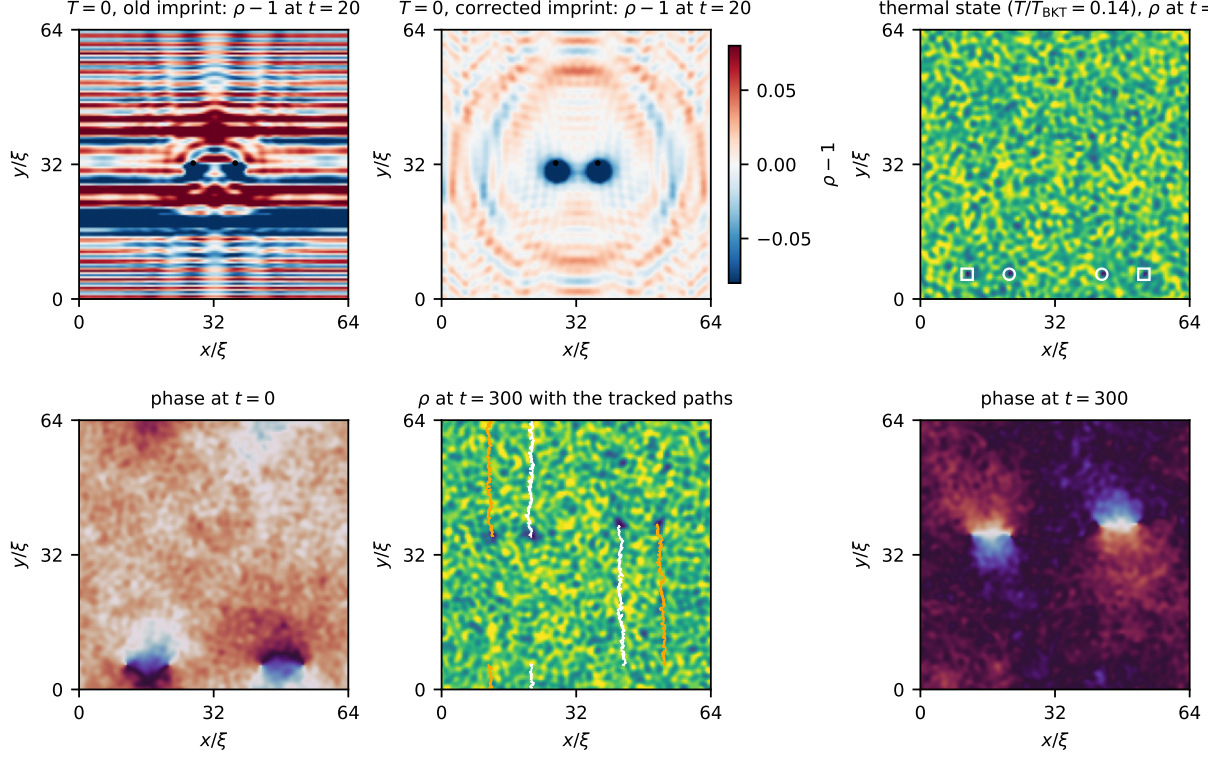


Figure 1: Top left and middle: density deviation $\rho - 1$ twenty time units after imprinting a single pair of separation 10 into a uniform condensate ($T = 0$, $L = 64$) with the imprint of the earlier rounds and with the corrected one. The old imprint leaves a phase step along a whole line of the box and a $1/r^2$ density tail, which turn into the sound sheet and the strong radiation on the left; the corrected one leaves the quiet wake on the right. Top right and bottom: a thermal state at $T/T_{\text{BKT}} = 0.14$ with two antiparallel pairs imprinted (density and phase at $t = 0$; density at $t = 300$ with the tracked trajectories of the four vortices, white for $+$, orange for $-$; phase at $t = 300$).

the customary per-vortex amplitude $[r^2/(r^2 + 2)]^{1/2}$ has a $1/r^2$ tail four times the quasi-static (Bernoulli) depletion, whose excess leaves the box as sound and pushes the vortices. We use instead the Bernoulli amplitude $[1 + |\mathbf{v}|^2/2]^{-1/2}$ built from the velocity of the imprinted phase itself. Figure 1 shows the two imprints side by side. With the old imprint a pair at $T = 0$ drifted by 0.6–0.8 in separation over 400 time units — an apparent friction as large as the thermal one at $0.14 T_{\text{BKT}}$ (cf. [16] on product-ansatz pairs); with the corrected one it does not move (below).

Geometry. A pair of separation d carries the impulse $2\pi\rho_s d$. In a periodic box that impulse cannot be shed without consequence (Section 4), so transport runs use two antiparallel pairs (a $+-$ pair at $x = L/4$ and a $-+$ pair at $x = 3L/4$, same separation, moving in opposite directions): zero net charge, zero net dipole moment, zero net impulse. Single pairs are used only for the question of Section 4.

Detection and tracking. Vortices are located by the phase winding around each plaquette (the rule proved correct in `VortexWinding.lean`) and refined to sub-grid precision by the zero of the least-squares plane through the four corner values of ψ : 0.01 at $T = 0$, with a thermal jitter of ≈ 0.3 at $T/T_{\text{BKT}} = 0.14$. The imprinted vortices are followed by continuity (same sign, nearest detection within 3.0 per time unit), positions are saved every time unit, and a track is used only while every vortex–antivortex distance exceeds 4ξ , below which a pair is a single

solitary wave rather than two vortices.

Estimators (identities of the model). Let H be the point-vortex Hamiltonian of the tracked configuration on the torus (the Weiss–McWilliams sum [17]). Equation (1) for all vortices is a skew part — the conservative flow, which also contains α' — plus $-\frac{\alpha}{2}\nabla H$, so that $\dot{H} = -\frac{\alpha}{2}|\nabla H|^2 = -2\alpha \sum_i |\mathbf{v}_{s,i}|^2$ for any configuration and any α' . Hence

$$\alpha_{\text{energy}} = -\frac{\Delta H}{2 \int \sum_i |\mathbf{v}_{s,i}|^2 dt}, \quad (2)$$

which needs no pairing of vortices and is blind to α' . Independently, the displacement of each vortex over a lag regressed on the two orthogonal predictors $\int \mathbf{v}_{s,i} dt$ and $-q_i \int \hat{z} \times \mathbf{v}_{s,i} dt$ gives $1 - \alpha'$ and α as the two coefficients; the mean square of the residuals of that regression against lag, $\langle |e|^2 \rangle = c_0 + 4\eta\tau$, gives the diffusion constant, which we quote only if the local scaling exponent on lags 20–400 lies in $[0.8, 1.2]$. For a stochastic pair with drift $-2\alpha\mathbf{d}/|\mathbf{d}|^2$ and isotropic diffusion $D = 2\eta$ of the separation, the Kosterlitz–Thouless pair distribution $|\mathbf{d}|^{-K}$, $K = 2\pi\rho_s/T$, carries zero probability current if and only if $DK = 2\alpha$, i.e. $\eta = \alpha T/(2\pi\rho_s)$: the Einstein statistic is $R_E = \eta K/\alpha$, equal to one when the pair gas can be in detailed balance at the measured stiffness. These statements — the dissipation identity, the d^2 law of a plane pair and its lifetime bound, the centre-velocity identity behind the α' estimator, and the zero-current characterisation of the Einstein relation — are theorems in Lean 4 (Appendix C): the proof assistant checks, from the model’s equations alone, that the quantities we compute from positions are what we say they are, with no hidden assumption.

Controls. The estimators recover known (α, α', η) from synthetic Langevin trajectories to 7%, 0.002 and 25%. At $T = 0$ (uniform condensate), with the corrected imprint, a pair of separation 9.8 neither shrinks nor diffuses over 400 time units ($|\alpha| \leq 4 \times 10^{-4}$, residual mean-square displacement bounded at 0.005), its members move at the torus point-vortex velocity to 0.14%, and a single pair’s field momentum is constant to 3×10^{-10} and equal to 0.96–0.99 of $2\pi nd$ — consistent with the theoretical expectation that an isolated pair does not shrink at $T = 0$ [18]. At $T/T_{\text{BKT}} = 0.14$ two antiparallel pairs give $\alpha = 0.00705 \pm 0.00012$, inside a window set before the run from [1]. Full gate record: Appendix A.

3 Friction, transverse coefficient and diffusion

Friction. Figure 2 (left) and Table 1 give α at the three temperatures from six runs each (pair sizes 8 and 12, three seeds, 2000 time units; eight runs at the coldest). The three criteria registered for it pass: the two pair sizes agree (ratios 0.91, 0.76, 1.04, within the registered $[0.7, 1.43]$), α increases with T , and the energy and regression estimators agree within 5, 13 and 18%. The warmest base contains a thermal vortex in more than 5% of its samples and its error is large; it is kept with that flag. What was not registered and is the cleanest statement of the data is the ratio $\alpha/(\rho_n/\rho) = 0.23, 0.26, 0.22$: within errors, **the friction is proportional to the normal fraction**, $\alpha \simeq 0.24 \rho_n/\rho$, over a factor 3.5 in ρ_n . At $0.14 T_{\text{BKT}}$ our 0.0062 is a factor two below the interpolation of [1], whose transition temperature is itself a rough estimate; at 0.27 and 0.43 no closed-field value existed, and ours lie in the range of the oblate-condensate measurements [19] and above the unitary-gas value [20]. An independent configuration of this programme (eight imprinted vortices at $d = 16, 0.14 T_{\text{BKT}}$) gives $\alpha = 0.0048\text{--}0.0057$ with the same estimators.

Transverse coefficient, and the frame in which it is defined. The regression gives $1 - \alpha'$ directly; Table 1 lists the values obtained in the frame of the simulation box. They are *negative*:

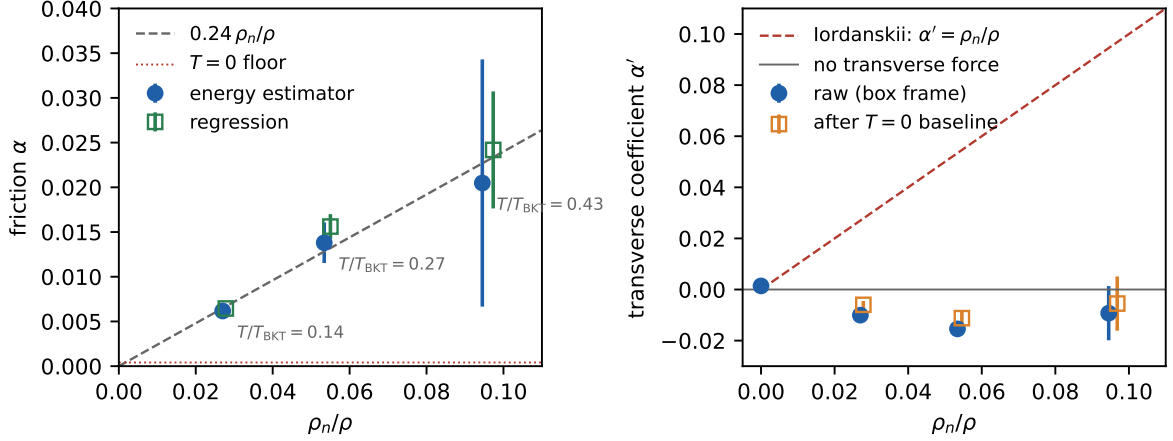


Figure 2: Left: friction α against the normal fraction ρ_n/ρ of the base state, from the energy estimator (2) and from the trajectory regression, with the line $\alpha = 0.24 \rho_n/\rho$ and the $T = 0$ floor of the instrument. Right: transverse coefficient α' as measured in the box frame (circles) and after subtracting the $T = 0$ kinematic baseline of a compressible pair at the sizes the fits use (squares), against the Iordanskii prediction $\alpha' = \rho_n/\rho$ and the no-transverse-force value 0. Errors are block jackknives over runs.

T/T_{BKT}	ρ_n/ρ	α (energy)	α (regression)	$\alpha(d_0=8)/\alpha(d_0=12)$	$\alpha/(\rho_n/\rho)$	α' (box frame)
0	0	4×10^{-4} (floor)	2×10^{-4}	—	—	+0.0014
0.14	0.027	0.0062 ± 0.0004	0.0065 ± 0.0004	0.91	0.23	-0.0100 ± 0.0015
0.27	0.053	0.0138 ± 0.0023	0.0156 ± 0.0014	0.76	0.26	-0.0154 ± 0.0029
0.43 [†]	0.094	0.021 ± 0.014	0.024 ± 0.007	1.04	0.22	-0.0093 ± 0.0106

Table 1: Transport coefficients of a vortex pair in the closed field ($L = 64$). [†]Base state with a thermal vortex in more than 5% of the samples.

the tracked vortices move 1–1.5% faster than the point-vortex velocity induced by their partners, while at $T = 0$ they move 0.14% slower. Neither theory has that sign, and the effect differs between the two pair sizes at the same temperature (-0.0135 ± 0.0018 at $d_0 = 8$, -0.0038 ± 0.0016 at 12), which points to a kinematic rather than a thermal origin. Two kinematic facts about a compressible pair on a torus must be accounted for before Eq. (1) is read, and both are measured, not assumed.

First, the velocity $\mathbf{v}_s$ in Eq. (1) is the *actual* superfluid velocity at the vortex, and on a torus that includes the uniform part of the flow. A configuration with net dipole moment $\mathbf{D} = \sum q \mathbf{r}$ carries the field momentum $2\pi n \hat{z} \times \mathbf{D}$, which in a periodic box is a mean superfluid velocity $\mathbf{u} = \mathbf{P}/(nL^2)$ of the whole fluid; the Weiss–McWilliams Green function, like any periodic point-vortex construction, fixes that uniform component by a gauge choice and not by the configuration’s momentum. The physical transverse force is a force exerted by the normal fluid, which in a closed box is at rest in the box frame, on a vortex moving relative to it; its signature is a difference between the vortex velocity and the *total* local superfluid velocity, $\mathbf{v}_s^{\text{WM}} + \mathbf{u}$. Omitting $\mathbf{u}$ makes a single pair appear to move faster than $\mathbf{v}_s$ by $|\mathbf{u}|d = 2\pi d^2/L^2$ — 6% at $d = 6$ and 49% at $d = 16$ in a $T = 0$ sweep of single pairs, and exactly what subtracting $\mathbf{u}$ removes (Table 2). For the antiparallel geometry $|\mathbf{u}| \leq 2 \times 10^{-3}$ and the two pairs move in opposite directions, so this correction to the production values is below 0.0002; it is decisive only for single pairs.

Second, a pair of finite size is not two point vortices: it is a solitary wave of the compressible field (the Jones–Roberts family), whose speed departs from $1/d$ by a correction of order $(\xi/d)^2$.

pair size d	3	4	6	8	10	16
measured speed / point-vortex speed, box frame	1.057	1.057	1.062	1.101	1.161	1.489
same, in the fluid frame ($\mathbf{v} - \mathbf{u}$)	≈ 1.04	≈ 1.03	1.010	1.004	1.004	1.005

Table 2: $T = 0$ kinematics of a single pair in the $L = 64$ box (uniform condensate, corrected imprint, 150 time units; $d = 3, 4$ from a 60-time-unit sweep). The box-frame excess is the mean flow $|\mathbf{u}| = 2\pi nd/(nL^2)$ of a configuration with net impulse; what remains in the fluid frame is the compressible correction to the pair speed.

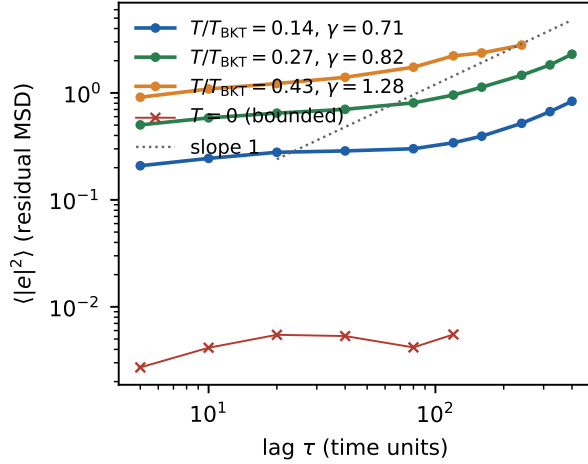


Figure 3: Mean-square residual displacement of a tracked vortex about the two-coefficient fit, against lag, at the three temperatures and at $T = 0$ (log-log). The local exponents over lags 20–400 are 0.71, 0.82 and 1.28; over lags ≥ 100 the two cooler temperatures give 1.1–1.2. The $T = 0$ curve is bounded (sound).

That correction is a property of the $T = 0$ field, independent of any normal fluid, so it must be measured at $T = 0$ and subtracted. In the fluid frame a $T = 0$ pair moves at the torus point-vortex speed to +0.4% for $d \geq 8$, +1% at $d = 6$ and +3–4% at $d = 3$ –4 (Table 2). The thermal fits of Table 1 use pairs of mean matched size 8.7–9.3 but weight the smaller, faster ones more, so their $T = 0$ baseline is $1 - \alpha'_{T=0} \approx 1.004$ to 1.01.

After both corrections the thermal transverse coefficient is $\alpha' = -0.006 \pm 0.002$, -0.011 ± 0.003 and -0.006 ± 0.011 at the three temperatures, with an uncertainty of the same order from the mapping between fit weights and pair sizes: α' is zero within about 1%, Figure 2 (right). The decision rule registered for this quantity was a weighted regression of α' on ρ_n/ρ including the $T = 0$ point; it gives slope $b = -0.31 \pm 0.05$ in the box frame (and a slope compatible with zero after the baseline), against the Iordanskii window $b \in [0.6, 1.4]$ and the no-transverse-force criterion “ 2σ upper bound on b below 0.3”. **The registered prediction — Iordanskii-like — is refuted at 18σ** ; the Iordanskii values +2.7%, +5.3% and +9.4% at the three temperatures are excluded, and the rule returns “no transverse force”. The eight-vortex configuration at $d = 16$ gives $1 - \alpha' = 1.0012$, consistent. We note that the measurement is of a classical Rayleigh–Jeans field at one cutoff; the phonon scattering that would produce an Iordanskii force in a quantum gas has its classical counterpart here, and that counterpart does not produce one at the 1% level in this regime.

Diffusion and the Einstein relation. Figure 3 shows the residual mean-square displacement against lag. At $T = 0$ it is bounded at 0.005: the residual is the vortex riding on the sound that remains in the periodic box. At $T > 0$ it has a short-lag plateau of 0.2–1.2 — the thermal jitter

of the detected core position, whose correlation time is that of the phonons — followed by growth. The local exponent on lags 20–400 is 0.71, 0.82 and 1.28 at the three temperatures; on lags ≥ 100 the two cooler ones are Brownian (1.1–1.2). Extracting a diffusion constant from a sub-diffusive window is ill-posed, which is why the rule registered before the analysis quotes η only where the exponent lies in $[0.8, 1.2]$: that admits the middle temperature alone, $\eta = (1.08 \pm 0.35) \times 10^{-3}$, giving $R_E = \eta K/\alpha = 2.1 \pm 0.7$; the two unquoted temperatures would give 3.2 and 1.4. **By the registered rule the Einstein relation is inconclusive** (two admitted temperatures were required). What the data say informally is that the vortex diffuses at 1.4–3 times the Einstein value, the excess decreasing with temperature — far from the hundredfold of [2], and not cleanly one. The sub-diffusion at the coldest temperature is a short-lag regime, not a bounded orbit: a finite quasi-periodic sum has a bounded mean-square increment, so growth refutes a regular island while a plateau never proves one (`QuasiPeriodicBound.lean`), and the spectrum of a stalled pair’s separation (next section) is a $1/f^{0.5}$ continuum, not lines. The autocorrelation of the residual velocity at $0.14 T_{\text{BKT}}$ has a small negative lobe, -0.010 ± 0.003 , at lag 79 — near the acoustic crossing time of the box, $L/c = 64$ — which is at least consistent with the vortex meeting its own shed sound. The physical reading we can defend is that at the lowest temperature the vortex’s bath is not yet Markovian on the lag scale of the fit: the kicks come from phonons whose period ($L/c \approx 64$) is not short compared with the lags, and the thermal vortices that make the warmest state Brownian also add drift.

4 A single pair in a closed box: the phonon wind

The impulse $2\pi\rho_s d$ of a shrinking pair has to go somewhere. In a periodic box it goes to the phonons, which then drift at $u = 2\pi\rho_s(d_0 - d)/(\rho_n L^2)$ in the direction of the pair’s motion, and the friction, which acts on the velocity of the vortex *relative to the normal fluid*, is reduced: $\dot{d} = -2\alpha(1/d - u)$. When $\pi\rho_s d_0^2 > 2\rho_n L^2$ the right-hand side has a zero at $b = [d_0 + (d_0^2 - 2\rho_n L^2/\pi\rho_s)^{1/2}]/2$ and the pair stalls there instead of annihilating (a theorem of the reduced equation, `wind_stall`). At $T/T_{\text{BKT}} = 0.14$, $L = 64$, a pair of $d_0 = 12$ should stall ($b = 10.2$ in the plane formula, ≈ 9 with the torus image forces) and a pair of $d_0 = 8$ should not; in an $L = 96$ box neither should.

Figures 4 and 5 show two single-pair runs of 4000 time units against the zero-impulse pairs of the same temperature. The zero-impulse pairs shrink steadily from 10 to 7–8 within 1500 time units; neither single pair annihilates. One shrinks from 11.6 to 9.3 by $t \approx 1800$ and stays there for the remaining 2200 time units (the late slope of d^2 is 4% of the early one); the other shrinks to 10.2 and re-expands to 11.2, moving back toward the predicted stall point. In both, the momentum carried by the modes with $|k| > 1$ follows the impulse shed by the pair — slopes 0.68 and 0.52 against the registered window $[0.5, 1.2]$ — and falls back when the pair re-expands. The object has a published precedent: in the truncated GPE a thermal cloud with net momentum relative to the condensate is a metastable counterflow decaying only by activated ring nucleation [21]; here the counterflow is generated by the pair itself. On the torus a single zero-circulation pair is a relative equilibrium of the point-vortex flow [22], so the wobble of the stalled pair is bath-driven; its spectrum over $t \in [2000, 4000]$ is a $1/f^{0.5}$ continuum with no dominant line.

The decisive control is the same pair in an $L = 96$ box at the same temperature ($T = 0.110$, $\rho_n/\rho = 0.029$, no thermal vortex), where $\rho_n L^2/(2\pi\rho_s) = 44 > d_0^2/4 = 36$ and the plane criterion predicts no stall. The registered reading of the control was binary — a pair that keeps shrinking supports the mechanism, a pair that stalls near the $L = 64$ separation refutes it — with a tie-break of two further runs if two runs disagree. [pending: W2 verdict on the four $L = 96$ runs: first two disagree (one keeps shrinking, late slope five times the early one; one shows a stall-like plateau near 9.6); runs 3 and 4 integrating] What the control shows beyond the binary reading is in Figure 4 (left): integrating the wind equation with the friction measured on the zero-impulse pairs, $\alpha = 0.0062$, and *no free parameter*, predicts at $L = 64$ a stall near 10 and at $L = 96$ not a

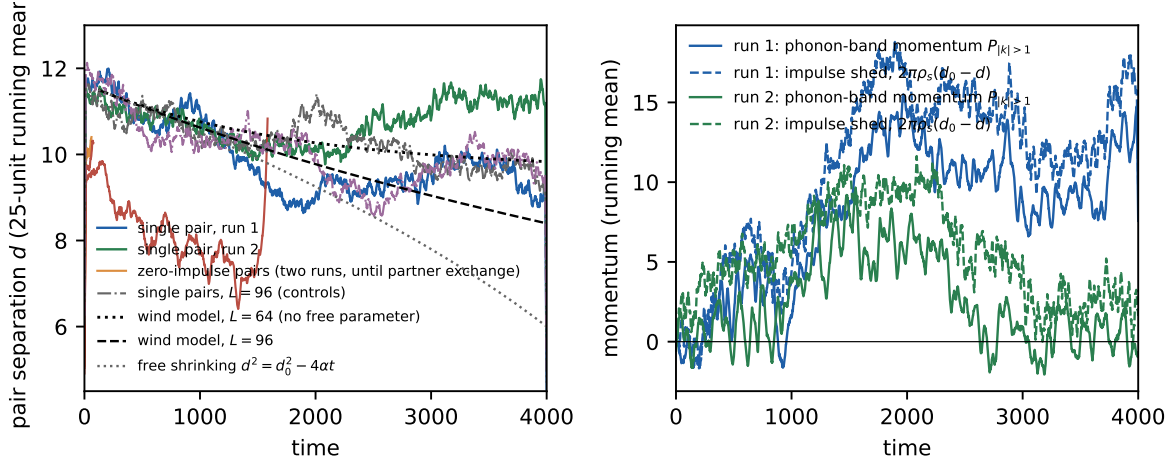


Figure 4: Left: separation of two single pairs ($d_0 = 12$, $T/T_{\text{BKT}} = 0.14$, $L = 64$, 4000 time units, 25-unit running mean), of the zero-impulse pairs of the same temperature until their partner exchange, and of the $L = 96$ control pairs (dash-dotted); the black curves are the wind equation integrated with the independently measured α for $L = 64$ (dotted) and $L = 96$ (dashed), the grey dotted curve is free shrinking. Right: the momentum of the modes with $|k| > 1$ projected on the pair’s initial impulse, against the impulse shed by the pair, $2\pi\rho_s(d_0 - d)$.

stall but a drive halved once the pair has shed two units of separation; the measured pairs end within about one unit of that curve at both sizes and three to five units above free shrinking ($d^2 = d_0^2 - 4\alpha t$), which would have taken them to 5–6. The suppression is graded with box size, as the mechanism says, and the binary criterion was too coarse an instrument for a graded effect; the stall map of the follow-up pre-registration (eleven cells in L , d_0 and T) is the quantitative test. If the mechanism holds, every friction measurement on a single pair in a closed box at low temperature is contaminated unless $\rho_n L^2 \gg \pi\rho_s d_0^2/2$ — a condition that the truncated-GPE measurements of the literature should be checked against, and the reason the production runs of Section 3 use a zero-impulse geometry.

5 The box-scale pair, equilibration, and the polarisability of the pair gas

One unscreened pair. In an earlier round, seven of eighteen large-box runs ($L = 128, 192$) reached a negative superfluid fraction with the same number of vortices as healthy partners at the same energy. A pre-registered test on their vortex positions refuted the natural suspicion of Onsager clustering: no like-sign clustering, positive vortex temperature, 98–99.7% of vortices paired with an opposite-sign neighbour. What distinguished them, found after the fact, was small and specific: the transverse current that point vortices drive, $(2\pi)^2 |\rho_q(k)|^2/k^2$ with $\rho_q(k) = \sum_j q_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}$, computed from the positions alone at the three longest wavelengths, tracks the measured transverse current across all eighteen runs with $r = 0.996$. The anomaly is one $O(1)$ unscreened pair of box size among hundreds of bound ones — the finite- k form of the polarisation term of [12, 13], and the surviving post-quench pair that Chu and Williams asked simulations to look for [23]. Its Boltzmann weight, $(L/\xi)^{-K} \lesssim 10^{-9}$, makes it a remnant of the initial relaxation, not an equilibrium feature; equilibrium finite-size stiffness is monotone in L [24, 25].

Equilibration. Coarsening under conservative dynamics has a dynamic exponent $z \approx 1.7$ –1.8 in this model [26] ($L^2 \ln L$ for dissipative XY dynamics [27, 28]; $L^2/\ln L$ for Hamiltonian XY [29]),

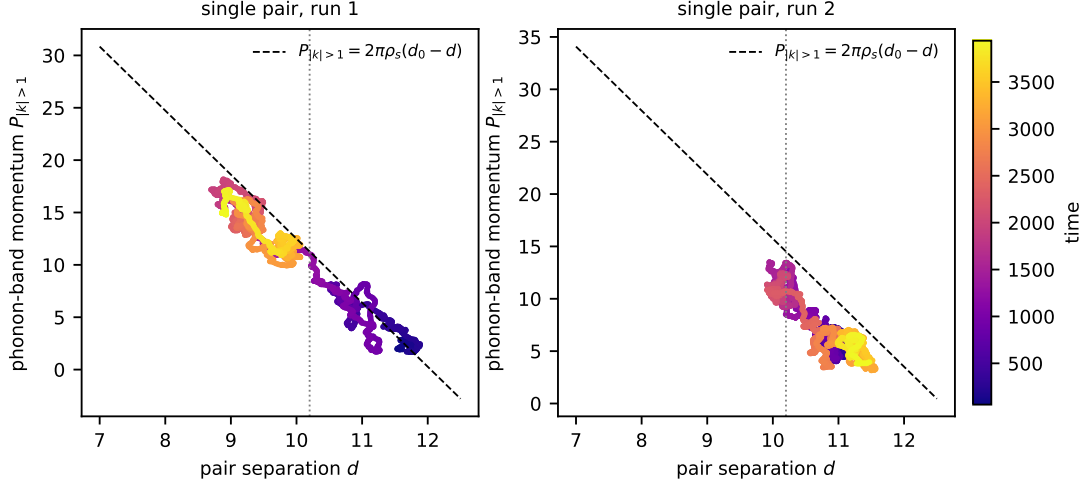


Figure 5: Phase portrait of the two stalled single pairs: phonon-band momentum against pair separation, coloured by time (50-unit running means). The dashed line is momentum conservation between the pair and the phonons, $P_{|k|>1} = 2\pi\rho_s(d_0 - d)$; the dotted line is the stall separation of the plane formula. Both trajectories move along the conservation line and settle near the predicted stall, the second drifting back along it.

so the warm-up that had cured the $L = 128$ states should scale to 11 000–14 600 time units at $L = 192$. We registered, before the runs, that continuing the six $L = 192$ trajectories to $t = 14\,500$ should admit at least five of six by the isotropy criterion, bring the vortex transverse response below 0.45 in every admitted run, and reduce it by at least a factor two in the two anomalous ones. Figure 6: all six are admitted (three were), the anomalous runs drop by factors 27 and 4.2, and five of six are below the threshold — the sixth, still at 0.80 with $K = 3.5$ against its partner’s 8.3, has not finished. The box-scale charge decays as the mechanism says in every run, and the equilibration time of an $L = 192$ box at this temperature exceeds 13 500 for some trajectories: longer than the L^2 and $z \approx 1.7$ estimates, as the $L^2 \ln L$ form would have it, on the evidence of one trajectory. On the equilibrated window the duality offsets $\eta_{g_1} K - 1$ are +0.20, +0.16 and +0.30 at the three energies: positive at every box size, so an earlier reading of these runs as showing a saturating finite-size offset rested on under-equilibrated states.

The Josephson offset. On the same equilibrated states the coherence exponent fitted with the exact torus Green function on $r \in [8, 32]$ gives $\eta_{g_1} K - 1 = +0.31 \pm 0.11$, unchanged by the fit window and equal to the naive-window value: the 12–27% excess over the Josephson relation reported at $L \leq 64$ is neither a geometry factor (cf. “close to unity” [30]) nor a finite-size transient. It persists at $L = 192$ in equilibrium. One candidate can already be excluded: that the offset is the signature of amplitude (density) fluctuations in $g_1 = \langle \psi^* \psi \rangle$. The phase-only correlator $\langle e^{i(\theta - \theta')} \rangle$, fitted on the same states, the same windows and the same torus Green function, carries *exactly* the same offset (+0.225 against +0.225 on $r \in [8, 48]$; difference 0.000 ± 0.001 over the six runs). The offset is not in the amplitude of g_1 . What remains is the estimator of K itself — the transverse current response at the lowest wavevectors of a fixed-momentum microcanonical field, for which no fluctuation formula has been derived — or the relation as applied at this coupling ($mg = 1$). Reported, not explained.

The dielectric relation that fails, and the polarisability that holds. We had registered, before the $L = 192$ states above existed, that the transverse current is, mode by mode and snapshot by snapshot, the point-vortex current of the detected charges plus a phonon part

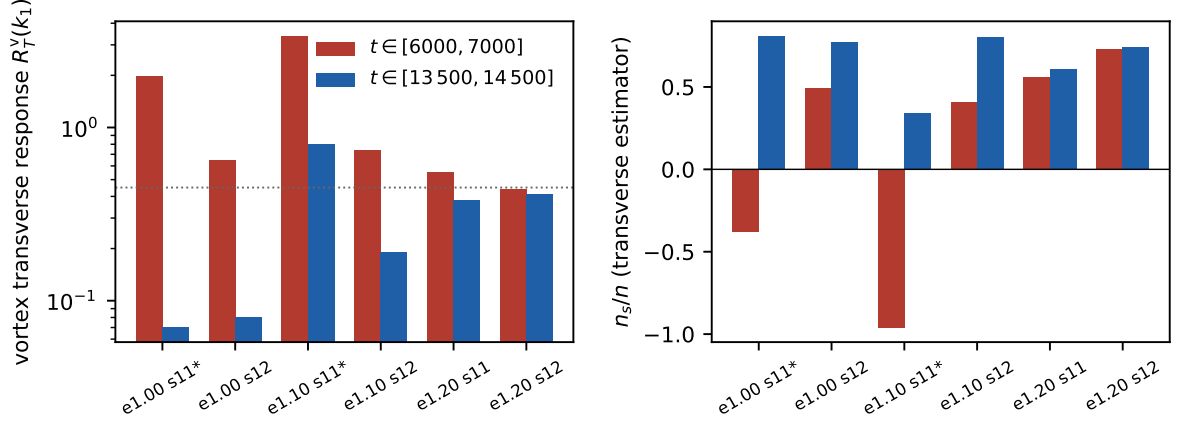


Figure 6: The six $L = 192$ trajectories at two sampling windows, $t \in [6000, 7000]$ and, after continuing them, $t \in [13500, 14500]$: the vortex-driven transverse response at the longest wavelength (left, logarithmic; the dotted line is the registered threshold 0.45) and the superfluid fraction from the transverse estimator (right). Starred runs are the two that had a negative superfluid fraction.

uncorrelated with them: a coherence $\gamma^2 \geq 0.4$ between $J_T(k, t)$ and $-2\pi i \rho_q(k, t)/|k|$ on every shell up to $|m|^2 = 16$ in at least five of six runs, with one real k -independent coefficient. Five controls on imprinted pair gases had failed on the amplitude's expected value (a tight pair carries less current than two point vortices, 0.92, 0.97, 0.99 at $d = 5, 8, 14$) while the coherence criterion passed in all of them — $\gamma^2 = 1.000$ at $T = 0$. On the six equilibrated thermal states, with 170–830 vortices almost all in tight pairs, **the coherence is 0.06–0.28**: the criterion fails in five of six, and the one run that passes is the one still carrying a box-scale pair. The phonon normal fraction, $f \approx 0.16$ –0.26, is the larger part of the transverse response there, and whatever the tight pairs contribute is not coherent with their instantaneous charge density at that wavevector. The $r = 0.996$ of the diagnosis above was, as the pre-registration itself warned, a correlation of run-averaged powers carried by the box-scale pairs. The diagnostic detects one unscreened pair; it is not a dielectric function of the pair gas.

What does hold is the static statement, Figure 7: in the five runs without a box-scale pair the charge structure factor $\langle |\rho_q(k)|^2 \rangle / k^2$ is independent of k out to $k\xi \approx 0.13$, and its value is the dipole polarisability $\Sigma_{\text{pairs}} d^2 / 2$ computed from the matched pair sizes, to 5–21% and always below it. This is the Kosterlitz–Thouless polarisability $\varepsilon - 1 \propto \Sigma d^2$ [10, 11] read directly from vortex positions in a Bose field, with a factor 0.8–0.95 that we read as the screening of dipoles by dipoles; the pair-size law itself had been measured before only in the XY model [31]. How much of the stiffness that polarisability accounts for is what the failure of the mode-by-mode relation leaves open: the current carried by a thermal pair of size 1–3 ξ is not the point-vortex current, and we do not yet have its form factor.

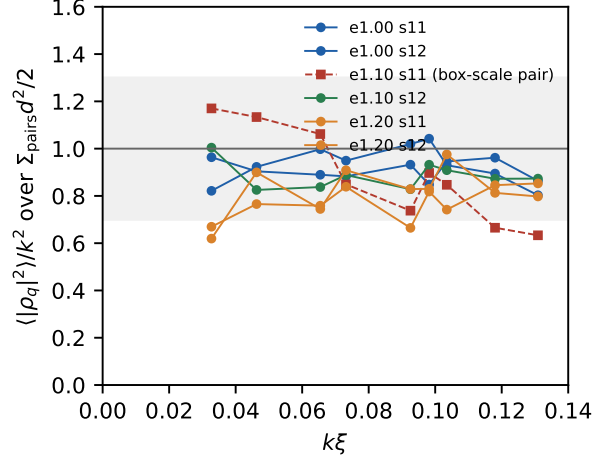


Figure 7: Vortex-charge structure factor of the six equilibrated $L = 192$ states, $\langle |\rho_q(k)|^2 \rangle / k^2$ on the shells $|m|^2 = 1\text{--}16$, divided by the dipole polarisability $\Sigma_{\text{pairs}} d^2 / 2$ computed from the minimum-cost matching of the detected vortices; the band is the registered $\pm 30\%$. The dashed run still carries a box-scale pair.

6 What this establishes, and what it does not

Established (pre-registered criteria passed, or controls stated as such).

- A vortex pair at $T = 0$ in the projected GPE neither shrinks nor diffuses and moves at the torus point-vortex velocity to 0.14% in the fluid frame; tight pairs ($d \leq 6$) move 1–4% faster, a compressible correction.
- $\alpha = 0.0062(4)$, $0.014(2)$, $0.02(1)$ at $T/T_{\text{BKT}} = 0.14$, 0.27 , 0.43 , independent of pair size and estimator; $\alpha \simeq 0.24 \rho_n / \rho$.
- α' is zero within about 1% after the $T = 0$ kinematics; the Iordanskii value is excluded at 18σ .
- The vortex diffuses at 1.4–3 times the Einstein value, not 100; the relation itself is undecided by our rule.
- The depressed stiffness of finite-box classical-field simulations is one unscreened box-scale pair, not clustering; given time growing faster than L^2 it decays (all six $L = 192$ runs admitted; one not finished).
- The thermal pair gas has a k -independent charge structure factor equal to the Kosterlitz–Thouless polarisability from the measured pair sizes to 5–21%.
- The Josephson offset $\eta_{g_1} K - 1 = +0.31 \pm 0.11$ survives at $L = 192$ in equilibrium, independent of window and torus geometry.
- The two imprint defects (phase seam, $1/r^2$ amplitude tail) and their correction.

Refuted, undecided, or open.

- *Refuted*: the registered Iordanskii-like prediction; the mode-by-mode dielectric relation in the thermal pair gas; the Onsager-cluster reading of the depressed-stiffness states; the earlier “saturating offset” reading.
- *Undecided by rule*: the Einstein relation (one admitted temperature).
- *Open*: the phonon-wind stall [pending: until the $L = 96$ control]; the cause of the Josephson offset; the current form factor of tight thermal pairs; why $\alpha \propto \rho_n$ with coefficient 0.24.
- *Scope*: a Rayleigh–Jeans field with one momentum cutoff, $mg = 1$, two dimensions; the classical-field normal fraction depends on the cutoff, so comparisons with quantum gases are made at equal ρ_n/ρ , not equal T/T_c .

On method. The point-vortex model is a variational projection of the field onto the manifold of vortex positions, and the coefficients measured here are the statistics of the field’s leak out of that manifold — the same object that a recent many-body experiment reads from the other end, by projecting a quantum state back onto a variational manifold at every step and asking how much coherence survives [32]. The hypotheses tested here were ranked by an automated five-minute-probe loop under priors fixed before any probe ran, from a local literature database of 774 entries whose 382 arXiv identifiers were verified against the arXiv API; the protocol, the gates that failed and the one verdict that was withdrawn are in Appendix A.

A Protocol, gates and amendments

Every pre-registration is a dated file in the repository (`docs/designs/`), committed before the data it governs existed; every change after a failed control is an amendment with its reason; every claim has a numbered entry in a ledger (`LEDGER.md`) whose status can be “withdrawn”. Table 3 is the record of this study.

control	outcome	consequence
G0 synthetic recovery	attempt 1 fail (generator let the two pairs exchange partners); attempt 2 pass	analysis keeps a track only while all opposite-sign distances ≥ 4
G1 $T = 0$ control	run 1 fail (pairs drift 0.6–0.8; single-pair momentum 0.71–0.91 of $2\pi nd$); run 2 pass	the two imprint defects found and corrected (Appendix B); an earlier known-answer gate built on the old imprint and a biased detector withdrawn
G2 thermal known answer	pass: $\alpha = 0.00705 \pm 0.00012$ in the window $[0.0028, 0.0112]$	production launched
Production, warm bases	first pass: all tracks lost inside the settling window	tracking radius $1.5 \rightarrow 3.0$ (nothing else); repeated
Dielectric D-G1, D-G1', D-G1'', D-G2', D-G2''	five fails on the amplitude’s expected value; coherence passed in all five	amplitude candidates reverted to the original three with no favourite; coherence and k -independence criteria unchanged
Equilibration E1/E2/E3	pass / fail (one run) / pass	reported as such

Table 3: Gate record. All data and the pre-registration texts are in the repository.

Decision rules, as registered. *Friction*: pair-size ratio in $[0.7, 1.43]$ at two of three temperatures;

monotone in T ; estimators within 25%. *Transverse coefficient*: weighted regression of α' on ρ_n/ρ with the $T = 0$ point; “Iordanskii-like” if the slope lies in $[0.6, 1.4]$ with intercept below 0.03, “no transverse force” if the 2σ upper bound on the slope is below 0.3; the registered prediction was Iordanskii-like. *Einstein*: η quoted only where the residual exponent on lags 20–400 lies in $[0.8, 1.2]$; “holds” if $R_E \in [0.5, 2]$ at every admitted temperature with at least two, “violated” if $R_E > 5$ or < 0.2 at two, inconclusive otherwise. *Wind*: W1, late d^2 slope below 25% of the early one in both single-pair runs (met in one by the letter, in both in substance); W2, no stall at $L = 96$; W3, phonon-band momentum against impulse shed with slope in $[0.5, 1.2]$ (met); supported only if all three hold. *Equilibration*: E1 admission $\geq 5/6$, E2 vortex response ≤ 0.45 in every admitted run, E3 $\geq 2\times$ drop in both anomalous runs. *Dielectric*: D1 coherence ≥ 0.4 on every shell in $\geq 5/6$ runs; D2 one real coefficient, k -ratio ≤ 1.25 ; D4 polarisability from matched pair sizes within 30%. One amendment (E-A1) was filed after the analysis it concerned and is labelled a retrodiction; one unit error in the D4 analysis code was caught and corrected before use.

B The corrected imprint

For vortices of charge $q_j = \pm 1$ at $\mathbf{r}_j = (x_j, y_j)$ in the square box of side L , the phase factor is built from the Jacobi function $\vartheta_1(u | \tau = i)$ with $u_j = \pi[(x - x_j) + i(y - y_j)]/L$:

$$\Phi_0(\mathbf{r}) = \prod_j \left(\frac{\vartheta_1(u_j)}{|\vartheta_1(u_j)|} \right)^{q_j}. \quad (3)$$

Under $y \rightarrow y + L$ each factor picks up $e^{-2iq_j\pi(x-x_j)/L}$, so for $\sum q_j = 0$ the product is periodic in y only up to the constant phase $2\pi D_x/L$, $D_x = \sum_j q_j x_j$ (and periodic in x). A single pair of separation d along x therefore carries a phase step $2\pi d/L$ along the whole line $y = 0$, which the projection turns into a sheet of sound. The corrected phase removes the jumps measured on Φ_0 itself, $\varphi_x = \arg\langle \Phi_0(\mathbf{r} + L\hat{x})\bar{\Phi}_0(\mathbf{r}) \rangle$ and likewise φ_y , by the uniform gradient (the torus counterflow of the configuration, in the minimal sector):

$$\Phi(\mathbf{r}) = \Phi_0(\mathbf{r}) e^{-i(\varphi_x x + \varphi_y y)/L}. \quad (4)$$

The amplitude is the quasi-static (Bernoulli) density of the imprinted flow, $\mathbf{v} = \nabla \arg \Phi$ evaluated by centred differences of Φ on the grid:

$$\psi \rightarrow \mathcal{P} \left[\psi \Phi \left(1 + \frac{1}{2} |\mathbf{v}|^2 \right)^{-1/2} \right] \times (\text{norm restored}). \quad (5)$$

This vanishes at the cores, approaches $1 - |\mathbf{v}|^2/2$ far from them, and for a neutral configuration removes a density that falls off as the pair’s own velocity field, $|\mathbf{v}|^2 \propto d^2/r^4$ — against the $1/r^2$ tail of the per-vortex factor $[r^2/(r^2 + 2)]^{1/2}$, which removes 21% of the density of a 64^2 box for 24 vortices before the norm is restored and launches the corresponding sound. Static check: a single pair of $d = 10$ has field momentum 60.2 against $2\pi nd = 62.8$ (was 49.8 with the seam) and peak density 1.007 (was 1.087). Dynamic check: Section 2, controls.

C Machine-checked statements

All modules are in `lean_src/`, checked by the Lean 4 kernel against Mathlib with the standard axioms only (`propext`, `Classical.choice`, `Quot.sound`). What is proved is that the estimators and the reduced equations used in the text are consequences of the model’s equations, not that the model describes the field.

- **DissipativeVortexDynamics**: for any flow $\dot{\mathbf{r}} = A(\nabla H) - \gamma \nabla H$ with A pointwise skew, $\frac{d}{dt}H = -\gamma|\nabla H|^2$ (the energy estimator, independent of the skew part and hence of α'); for the plane pair, $|\mathbf{d}|^2 = |\mathbf{d}_0|^2 - 4\alpha t$ on the interval of existence, the lifetime bound, and the centre velocity $(1 - \alpha')/|\mathbf{d}|$ perpendicular to the axis; for $\dot{d} = -c(d - a)(d - b)/d$ with $0 < a < b$, a pair starting at or above b never falls below it (the wind stall).
- **EinsteinRelation**: the pair distribution $|\mathbf{d}|^{-K}$ carries zero Fokker–Planck current for drift $-2\alpha\mathbf{d}/|\mathbf{d}|^2$ and diffusion D iff $DK = 2\alpha$; in vortex units $\eta = \alpha T/(2\pi\rho)$.
- **PairPolarisation**: $\|\rho_q(k)\| \leq \|k\| \sum_i \|\mathbf{d}_i\|$; $\rho_q(k) = i\mathbf{k} \cdot \mathbf{P}(k)$ with remainder $\leq \sum_i (\mathbf{k} \cdot \mathbf{d}_i)^2$ when $|\mathbf{k} \cdot \mathbf{d}_i| \leq 1$; a k -independent ceiling of the bound-pair transverse response.
- **MatchingScreening**: $\|\rho_q(k)\| \leq \|k\| \times (\text{the length of any vortex–antivortex matching})$, on the plane and the torus; the Fourier velocity of point vortices is transverse with $\|\hat{v}\|^2 = (2\pi)^2 \|\rho_q\|^2 / \|k\|^2$.
- **KTFlow**: the exact invariant $2u - \pi \ln u - 2\pi^3 y^2$ of the Kosterlitz flow and the trapping of the superfluid side at $K = 2/\pi$.
- **QuasiPeriodicBound**: increments of a finite trigonometric sum are bounded by $2 \sum |a_k|$, so a growing mean-square displacement excludes a quasi-periodic orbit.
- **VortexWinding, TopologicalProtection**: the plaquette winding rule and the integer conservation of winding, from earlier rounds.

D Code

The field solver, the instrument and the estimators are Python (NumPy) in the repository (exploration/pgpe/: pgpe.py, vortex_transport.py, transport_estimators.py, dielectric_modes.py); a Rust port of the instrument is in rusty-SUNDIALS (qf-pgpe, example vortex_transport.rs), cross-checked against the Python one to 5×10^{-7} in positions. The two pieces a reader needs to reproduce the measurements are short.

The corrected imprint (Appendix B; theta1 is the Jacobi function with nome $e^{-\pi}$, $\mathbf{s}$ the solver with grid $\mathbf{s}.N$, box $\mathbf{s}.L$, projector $\mathbf{s}.P$):

```
def imprint_v2(s, c, pos, q):
    x = np.arange(s.N) * s.dx; X, Y = np.meshgrid(x, x, indexing="ij"); Z = X + 1j * Y
    def factor(Zs):
        ph = np.ones(Zs.shape, complex)
        for (xj, yj), qj in zip(pos, q):
            t = theta1(np.pi * (Zs - (xj + 1j * yj)) / s.L); u = t / (np.abs(t) + 1e-300)
            ph = ph * (u if qj > 0 else np.conj(u))
        return ph
    ph = factor(Z)
    jx = np.angle(np.mean(factor(Z + s.L) * np.conj(ph)))
    jy = np.angle(np.mean(factor(Z + 1j * s.L) * np.conj(ph)))
    ph = ph * np.exp(-1j * (jx * X + jy * Y) / s.L)
    vx = np.angle(np.roll(ph, -1, 0) * np.conj(np.roll(ph, 1, 0))) / (2 * s.dx)
    vy = np.angle(np.roll(ph, -1, 1) * np.conj(np.roll(ph, 1, 1))) / (2 * s.dx)
    amp = np.sqrt(1.0 / (1.0 + 0.5 * (vx**2 + vy**2)))
    c2 = s.modes(s.psi(c) * ph * amp)
    return c2 * np.sqrt(s.norm(c) / s.norm(c2))
```

The estimators (pv_velocity returns the torus point-vortex velocity at every vortex, pv_energy the Weiss–McWilliams energy; U are the unwrapped tracked positions, shape (time, vortex, 2)):

```
V = np.array([pv_velocity(np.mod(U[k], L), q, L) for k in range(nt)])
H = np.array([pv_energy(np.mod(U[k], L), q, L) for k in range(nt)])
```

```

S = (V**2).sum(axis=(1, 2)) # sum_i |v_s,i|^2
I = np.concatenate([[0.0], np.cumsum(0.5 * (S[1:] + S[:-1]) * np.diff(t))])
alpha_energy = -np.polyfit(2 * I, H, 1)[0] # dH/dt = -2 alpha sum |v_s|^2
Bv = -q[None, :, None] * np.stack([-V[...], 1], V[...], 0]), -1) # -q z x v_s
CA = np.concatenate([np.zeros_like(V[1:]), np.cumsum(0.5*(V[1:]+V[:-1])*dt, 0)])
CB = np.concatenate([np.zeros_like(V[1:]), np.cumsum(0.5*(Bv[1:]+Bv[:-1])*dt, 0)])
D, A, B = U[lag:] - U[:-lag], CA[lag:] - CA[:-lag], CB[lag:] - CB[:-lag]
M = [[(A*A).sum(), (A*B).sum()], [(A*B).sum(), (B*B).sum()]]
one_minus_alpha_prime, alpha_regression = np.linalg.solve(M, [(A*D).sum(), (B*D).sum()])
e = D - one_minus_alpha_prime * A - alpha_regression * B # residuals -> <|e|^2>(lag) = c0
+ 4 eta lag

```

References

- [1] Vishwanath Shukla, Marc Brachet, and Rahul Pandit. Superfluid mutual-friction coefficients from vortex dynamics in the two-dimensional galerkin-truncated gross-pitaevskii equation. 2014. arXiv:1412.0706.
- [2] Tyler W. Neely, Guillaume Gauthier, Charles Glasspool, Matthew J. Davis, and Matthew T. Reeves. Melting of a vortex matter wigner crystal. 2024. arXiv:2402.09920.
- [3] S. V. Iordanskii. On the mutual frictions between the normal and superfluid components in a rotating bose gas. *Annals of Physics* 29, 335-349, 1964.
- [4] E. B. Sonin. Magnus force in superfluids and superconductors. *Phys. Rev. B* 55, 485, 1997. arXiv:cond-mat/9606099.
- [5] D. J. Thouless, Ping Ao, and Qian Niu. Transverse force on a quantized vortex in a superfluid. *Phys. Rev. Lett.* 76, 3758, 1996. arXiv:cond-mat/9603196.
- [6] Zain Mehdi, Joseph J. Hope, Stuart S. Szigeti, and Ashton S. Bradley. Mutual friction and diffusion of two-dimensional quantum vortices. *Phys. Rev. Research* 5, 013184, 2023. arXiv:2205.04065.
- [7] Giorgio Krstulovic and Marc E. Brachet. Kelvin waves, mutual friction, and fluctuations in the gross-pitaevskii model. *J. Low Temp. Phys.* 212, 321, 2023. arXiv:2305.00236.
- [8] Vinay Ambegaokar, B. I. Halperin, David R. Nelson, and Eric D. Siggia. Dynamics of superfluid films. *Phys. Rev. B* 21, 1806, 1980.
- [9] Yuri A. Sergeev. Mutual friction in bosonic superfluids: A review. *J. Low Temp. Phys.* 212, 251-305, 2023.
- [10] J. M. Kosterlitz and D. J. Thouless. Ordering, metastability and phase transitions in two-dimensional systems. *J. Phys. C: Solid State Phys.* 6, 1181-1203, 1973.
- [11] J. M. Kosterlitz. The critical properties of the two-dimensional xy model. *J. Phys. C: Solid State Phys.* 7, 1046-1060, 1974.
- [12] A. Vallat and H. Beck. Coulomb-gas representation of the two-dimensional xy model on a torus. *Phys. Rev. B* 50, 4015-4029, 1994.
- [13] Michael F. Faulkner, Steven T. Bramwell, and Peter C. W. Holdsworth. An electric-field representation of the harmonic xy model. *J. Phys.: Condens. Matter* 29, 085402, 2017. arXiv:1610.06692.
- [14] Christopher J. Foster, P. Blair Blakie, and Matthew J. Davis. Vortex pairing in two-dimensional bose gases. *Phys. Rev. A* 81, 023623, 2010. arXiv:0912.1675.

- [15] Krzysztof Gawryluk and Mirosław Brewczyk. Signatures of a universal jump in the superfluid density in two-dimensional bose gas with finite number of particles. *Phys. Rev. A* **99**, 033615, 2019. arXiv:1809.10967.
- [16] Cecilia Rorai, K. R. Sreenivasan, and Michael E. Fisher. Propagating and annihilating vortex dipoles in the gross-pitaevskii equation. *Phys. Rev. B* **88**, 134522, 2013. arXiv:1212.6389.
- [17] Jeffrey B. Weiss and James C. McWilliams. Nonergodicity of point vortices. *Phys. Fluids A* **3**, 835-844, 1991.
- [18] Andrew Lucas and Piotr Surówka. Sound-induced vortex interactions in a zero-temperature two-dimensional superfluid. *Phys. Rev. A* **90**, 053617, 2014. arXiv:1408.5913.
- [19] Geol Moon, Woo Jin Kwon, Hyunjik Lee, and Yong il Shin. Thermal friction on quantum vortices in a bose-einstein condensate. *Phys. Rev. A* **92**, 051601(R), 2015. arXiv:1509.02236.
- [20] W. J. Kwon, G. Del Pace, K. Xhani, L. Galantucci, A. Muzi Falconi, M. Inguscio, F. Scazza, and G. Roati. Sound emission and annihilations in a programmable quantum vortex collider. *Nature* **600**, 64-69, 2021. arXiv:2105.15180.
- [21] Giorgio Krstulovic and Marc Brachet. Energy cascade with small-scales thermalization, counterflow metastability and anomalous velocity of vortex rings in fourier-truncated gross-pitaevskii equation. *Phys. Rev. E* **83**, 066311, 2011. arXiv:1010.0116.
- [22] Klas Modin and Milo Viviani. Integrability of point-vortex dynamics via symplectic reduction: a survey. *Arnold Math. J.* **7**, 357-385, 2020. arXiv:2003.00716.
- [23] Han-Ching Chu and Gary A. Williams. Quenched kosterlitz-thouless superfluid transitions. *Phys. Rev. Lett.* **86**, 2585, 2001. arXiv:cond-mat/0010461.
- [24] Nikolay Prokof'ev and Boris Svistunov. Two-dimensional weakly interacting bose gas in the fluctuation region. *Phys. Rev. A* **66**, 043608, 2002. arXiv:cond-mat/0206223.
- [25] Martin Hasenbusch. The two dimensional xy model at the transition temperature: A high precision monte carlo study. *J. Phys. A* **38**, 5869, 2005. arXiv:cond-mat/0502556.
- [26] Andrew J. Groszek and Thomas P. Billam. Universal coarsening of a two-dimensional bose gas under conservative evolution. 2026. arXiv:2601.02687.
- [27] A. J. Bray, A. J. Briant, and D. K. Jervis. Breakdown of scaling in the nonequilibrium critical dynamics of the two-dimensional xy model. *Phys. Rev. Lett.* **84**, 1503, 2000. arXiv:cond-mat/9902362.
- [28] Asja Jelić and Leticia F. Cugliandolo. Quench dynamics of the 2d xy model. *J. Stat. Mech.*, 2011. arXiv:1012.0417.
- [29] Keekwon Nam, Woon-Bo Baek, Bongsoo Kim, and Sung Jong Lee. Coarsening of two dimensional xy model with hamiltonian dynamics: Logarithmically divergent vortex mobility. *J. Stat. Mech.*, 2012. arXiv:1211.1462.
- [30] Markus Holzmann and Gordon Baym. Condensate superfluidity and infrared structure of the single-particle green's function: The josephson relation. *Phys. Rev. B* **76**, 092502, 2007. arXiv:cond-mat/0703755.
- [31] Mingyu Fan, Gary A. Williams, Karla Galdamez, and Charlie McDowell. Coulomb gas sum rules for vortex-pair fluctuations in 2d superfluids. 2022. arXiv:2205.06371.

- [32] Hang Dong, Jie Ren, Andrew Hallam, Han Wang, Zhengyi Cui, Yiren Zou, Junlin Wang, Hekang Li, Qiujiang Guo, Zhen Wang, Lei Ying, and Zlatko Papic. Quantum many-body mixed phase space revealed by hybrid feedback control. *Nature Physics*, 2026.